OFDM Simulation and Performance Evaluation

Almansur Kakimov

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Abstract

The project uses computer simulations to undertake a detailed look into orthogonal frequency division multiplexing (OFDM). The study examines a variety of factors that influence OFDM performance, including bit error rate (BER), signal-to-jammer ratio (SJR), and modulation techniques. The research also examines the use of Multiple-Input Multiple-Output (MIMO) in OFDM, the effects of adaptive modulation, and how reinforcement learning (Q-learning) might boost performance. Simulations in a controlled environment were performed to examine system robustness under various channel conditions using Python-based numerical analysis.

1 Introduction

Orthogonal Frequency Division Multiplexing (OFDM) is a prominent modulation scheme in modern wireless communications because of its resistance to multipath fading and excellent spectral efficiency. It serves as the foundation for several communication technologies, such as LTE, Wi-Fi (802.11), and 5G. This study looks into the theoretical basis, mathematical models, and computer simulations of OFDM systems, particularly MIMO-OFDM combinations. Furthermore, the study investigates the role of Q-learning in optimizing system features, with a focus on dynamic spectrum allocation and adaptive modulation for improved dependability in noisy situations.

2 Theoretical Background

2.1 OFDM Signal Representation

OFDM Signal Representation OFDM separates a broadband channel into numerous narrowband subcarriers that are orthogonal to one another. The mathematical representation of an OFDM signal is:

$$X(k) = \sum_{n=0}^{N-1} x(n) e^{-j2\pi kn/N} \tag{1}$$

where:

- $X(k)$ represents the frequency domain signal,
- $x(n)$ is the time-domain input,
- N is the total number of subcarriers.

The corresponding time-domain signal is obtained using the Inverse Fast Fourier Transform (IFFT):

$$x(n) = \frac{1}{N} \sum_{k=0}^{N-1} X(k) e^{j2\pi kn/N} \tag{2}$$

A Cyclic Prefix (CP) is added to counteract inter-symbol interference (ISI), ensuring robustness against multipath fading.

2.2 MIMO-OFDM Systems

MIMO-OFDM extends OFDM by incorporating multiple antennas at the transmitter and receiver, significantly improving spectral efficiency and reliability. The received signal at the receiver in a MIMO-OFDM system is modeled as:

$$Y = HX + N \tag{3}$$

where:

- Y is the received signal vector,
- H is the channel matrix,
- X is the transmitted signal vector,
- N is the noise component.

Detection methods such as Zero-Forcing (ZF) and Minimum Mean Square Error (MMSE) are applied to recover the transmitted symbols efficiently.

2.3 Adaptive Modulation in OFDM

To improve system performance, adaptive modulation dynamically alters the modulation order in response to channel conditions. The system chooses amongst BPSK, QPSK, 16-QAM, and 64-QAM to maximize data rate while preserving a reasonable BER. The switching threshold is given by:

$$SNR_{threshold} = \frac{E_b}{N_0} \times BER_{target} \tag{4}$$

3 Methodology

The simulations were implemented using Python with the following key steps:

- **Parameter Initialization**: Setting modulation order, number of subcarriers, cyclic prefix length, and SNR range.
- **OFDM Transmitter**: Signal generation, modulation, IFFT application, and cyclic prefix addition.
- **Channel Modeling**: Simulation of Additive White Gaussian Noise (AWGN), Rayleigh fading, and jamming attacks.
- **MIMO Detection**: Implementation of ZF and MMSE equalization for MIMO-OFDM.
- **Machine Learning Enhancement**: Q-learning for adaptive modulation and power control.
- **Performance Metrics**: BER computation, throughput evaluation, and signal constellation visualization.

4 Q-Learning for OFDM Optimization

Q-learning is used to dynamically optimize the modulation order and power allocation. The Q-learning update equation is:

$$Q(s,a) = Q(s,a) + \alpha[r + \gamma \max_a Q(s',a) - Q(s,a)] \tag{5}$$

where:

- s is the current state (channel SNR, interference levels),
- a is the chosen action (modulation type, power level),
- r is the reward function based on BER and throughput,
- γ is the discount factor,
- α is the learning rate.

The agent explores different modulation and power configurations, learning an optimal strategy through iterative updates.

5 Results and Discussion

- **BER vs. SNR**: As expected, higher SNR leads to lower BER. Adaptive modulation outperforms fixed modulation schemes.

- **MIMO-OFDM Gain**: MIMO-OFDM provides a significant improvement in spectral efficiency, especially in high-mobility scenarios.
- **Impact of Jamming**: The system's robustness to jamming is evaluated by introducing interference signals and observing BER variations.
- **Effect of Q-Learning**: The reinforcement learning model minimizes BER while increasing throughput by dynamically altering modulation levels and power distribution.

6 Conclusion

This paper provides insight into OFDM system performance, highlighting the benefits of adaptive modulation, MIMO integration, and reinforcement learning-based optimization. The application of Q-learning enables dynamic adaptation, which improves dependability in actual contexts with noise and interference.

7 Future Work

Future research may explore:

- Deep Reinforcement Learning (DQN, PPO) for advanced OFDM parameter optimization.
- Hybrid MIMO-NOMA (Non-Orthogonal Multiple Access) systems.

8 References

References

[1] U. Altun and E. Basar, "A Reinforcement Learning-Assisted OFDM-IM Communication System against Reactive Jammers," *IEEE Transactions on Cognitive Communications and Networking*, pp. 1–1, 2024. doi: 10.1109/tccn.2024.3522092.